

Reliable Classification of High Explosive and Chemical/Biological Artillery Using Acoustic Sensors

Sachi Desai¹, Myron E. Hohil¹, Henry Bass² and Jim Chambers²

¹US Army RDECOM, Picatinny Arsenal, NJ 07806

Email: mhohil@pica.army.mil; sdesai@pica.army.mil;

²National Center for Physical Acoustics, University, MS 38677

Email: pabass@olemiss.edu; chambers@olemiss.edu;

Abstract

Feature extraction methods based on the discrete wavelet transform and multiresolution analysis are used to develop a robust classification algorithm that reliably discriminates between conventional and simulated chemical/biological artillery rounds via acoustic signals produced during detonation utilizing a generic acoustic sensor. Based on the transient properties of the signature blast distinct characteristics arise within the different acoustic signatures because high explosive warheads emphasize concussive and shrapnel effects, while chemical/biological warheads are designed to disperse their contents over large areas, therefore employing a slower burning, less intense explosive to mix and spread their contents. The ensuing blast waves are readily characterized by variations in the corresponding peak pressure and rise time of the blast, differences in the ratio of positive pressure amplitude to the negative amplitude, and variations in the overall duration of the resulting waveform. Unique attributes can also be identified that depend upon the properties of the gun tube, projectile speed at the muzzle, and the explosive burn rates of the warhead. The algorithm enables robust classification of various airburst signatures using acoustics. It is capable of being integrated within an existing chemical/biological sensor, a stand-alone generic sensor, or a part of a disparate sensor suite. When emplaced in high-threat areas, this added capability would further provide field personnel with advanced battlefield knowledge without the aid of so-called "sniffer" sensors that rely upon air particle information based on direct contact with possible contaminated air. In this work, the discrete wavelet transform is used to extract the predominant components of these characteristics from air burst signatures at ranges exceeding 2km while maintaining temporal sequence of the data to keep relevance to the transient differences of the airburst signatures. Highly reliable discrimination is achieved with a feedforward neural network classifier trained on a feature space derived from the distribution of wavelet coefficients and higher frequency details found within different levels of the multiresolution decomposition the neural network then is capable of classifying new airburst signatures as Chemical/Biological or High Explosive.

Keywords: Acoustic Sensors, CBRN, Chemical, Biological, Neural Networks, Wavelets, Multiresolutional Analysis

1. INTRODUCTION

The Army is currently developing acoustic overwatch sensor systems that will provide extended range surveillance, detection, and identification for force protection and tactical security on the battlefield. A network of such sensors remotely deployed in conjunction with a central processing node (or gateway) will provide early warning and assessment of enemy threats, near real-time situational awareness to commanders, and may reduce potential hazards to

the soldier. In contrast, the current detection of chemical/biological (CB) agents expelled into a battlefield environment is limited to the response of chemical sensors that must be located within close proximity to the CB agent. Since chemical sensors detect hazardous agents through contact, the sensor range to an airburst is the key-limiting factor in identifying a potential CB weapon attack. The associated sensor reporting latencies must be minimized to give sufficient preparation time to field commanders, who must assess if an attack is about to occur, has occurred, or if occurred, the type of agent that soldiers might be exposed to. The long-range propagation of acoustic blast waves from heavy artillery blasts, which are typical in a battlefield environment, introduces a feature for using acoustics and other disparate sensor technologies for the early detection and identification of CB threats. Employing disparate sensor technologies implies that warning of a potential CB attack can be provided to the soldier more rapidly and from a safer distance when compared to that which conventional methods allow. This capability facilitates the necessity of classifying the types of rounds that have burst in a specified region in order to give both warning and provide identification of CB agents found in the area. The added information integrated into the common operating picture will help field commanders (more) accurately plan and execute measures that must be taken in order to ensure the proper precautions to a mission [10]. Such a warning system would also help reduce the time-consuming, manpower intensive and dangerous tasks associated with identifying the airburst, allowing a field commander to make rapid and accurate judgments that insure greater safety and lessen exposure for the soldiers.

This paper supports acoustic sensor technologies to distinguish among conventional, i.e. high explosive (HE), and CB artillery blasts using characteristics that remain discernible with long-range wave propagation and degradation, firing time and detonation (either air or ground). The outcome illustrated herein is based on an analysis conducted using acoustic signature data collected during two independent data collection exercises. The principal set of artillery blast data was collected at YUMA Proving Ground on 11-12 August 1995 in a field test experiment conducted by the National Center of Physical Acoustics NCPA at the University of Mississippi in cooperation with ARDEC (Picatinny Arsenal). The reason for the experiment was to examine whether CB rounds could be discriminated from HE rounds via acoustic and seismic sensor technologies. The field test included the detonation and/or airburst of three categories of explosive rounds; two categories of artillery rounds considered close analogs to chemical and biological warheads, and HE rounds most frequently used in the battlefield. The data set comprised of 8 ground impacts from HE rounds, 23 ground impacts from a Type A CB round and 8 air-bursts from a Type B CB round. The data was collected at seven sensors sites as shown in Figure 1. The rounds were fired from a self-propelled howitzer situated several kilometers to the west of a linear array of 5 sensor locations designated along an axis offset from the firing radial by 700m. Sensor site S3 coincides with the ordinate axis of the estimated projectile impact point at 5km from the firing position. Microphones were emplaced at sensor sites S2, S4, S1 and S5 at a distance of 500m, 600m, 1000m and 1200m from S3, respectively. Geophones accompanied the microphones at positions S3 and S5, and a flash detector was emplaced at position S6, 2000 meters from S3. An acoustic transducer was positioned at site S7, 400m from S3 and in line with the estimated projectile impact point.

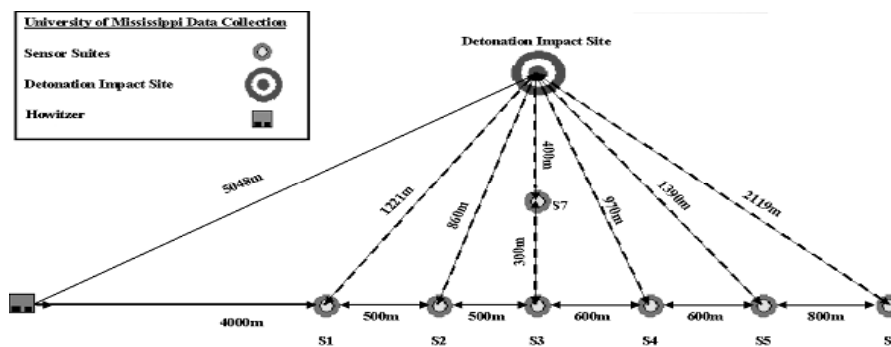


Figure 1 – Sensor Deployment at Yuma Proving Grounds

A preliminary investigation conducted by researchers at the National Center for Physical Acoustics exposed distinctive differences between airburst signatures from the three types of rounds, and a subsequent study identified features within the blast waves that could be used to discriminate between HE and CB blasts acoustically [1]. In this study, the rapid ignition of the HE rounds was found to generate larger amplitude, shorter duration signatures in contrast to the longer duration signals representative of the slower burning phosphorus, ignited in the Type A shell upon ground

impact. It was determined that, if previous information regarding range to the blast was understood, the amplitude alone would be adequate to identify a CB round from its conventional HE counterpart. Discriminating attributes such as high frequency precursor signals produced by the passing of supersonic shrapnel elements from the exterior of HE rounds were identified prior to the main blast wave and noticeable at ranges up to 1 km. Signatures for the Type B round illustrated a short duration pulse that precedes the slow burn of chemical agents, an aspect resulting from the ejection of the round from the canister.

A second set of experimental data was collected at Dugway Proving Ground (DPG). The intention of this exercise was to examine if acoustic, seismic, and imagery sensors could be used to differentiate among conventional HE projectiles and those filled with either chemical and/or biological agents [2]. Two hundred and sixty rounds were fired, approximately half of which were fused to detonate on impact, while the other half were fused to detonate in the air. The HE rounds contained a larger amount of explosives when compared to the simulated chemical rounds that were developed to open at detonation and release a fill consisting of polyethylene glycol and water. The data examined in this paper was collected from three sensor sites arranged in a triangle and numbered with indication of the range from the detonation zone. Site 1 and site 3 were situated on the same radial line from the expected detonation area at ranges of 540m and 1665m respectively. Site 2 was positioned 1026 m radially outward and 514 m to the left from the detonation area.

The latter field test experiment has begun numerous studies that have created a wealth of research concerning the discrimination of HE and CB rounds using disparate sensor technologies [2-8]. In [7], characteristics found in the seismic signatures were identified to provide themselves well toward the discrimination of HE and CB artillery blasts, particularly in the case of the point detonation trials. This work applied time and frequency domain approaches to obtain features useful in the discrimination between HE and CB rounds. Wavelets were employed to de-noise the non-stationary signals and recreate the time domain information for use in discovering robust features. Explosion start and stop times were calculated and exploited to find the qualities found in the envelope information. Time domain discriminating features were identified, including the shrapnel noise reported in [1], and frequency domain features such as the instantaneous and maximum bandwidth. Acoustic features were also identified and presented in collaboration between the Edgewood Chemical Biological Center (ECBC) and Army Research Lab (ARL), in combination with seismic research conducted in [7] (c.f. [2-8]).

The most prominent characteristics within each type of blast are readily identified in Figure 2; the high frequency precursors from the HE blast prior to the main blasts as shown in Figure 2a, the comparatively slow burn and weak under pressure associated with the Type A CB blast in Figure 2b, and a short duration pulse that precedes the slow burn of agents within the Type B Base ejection rounds shown in Figure 2c. In here, we identify range invariant features found within the multiresolution decomposition of the corresponding signatures in an attempt to further promote disparate sensor technology for CB/HE discrimination. The subsequent growth of a feature space representation for classification is based upon data acquired during separate data collection exercises, which when combined, introduce variations within the intraclass and interclass attributes for the HE and Type A CB classes of artillery blasts. This is principally due to the different types of payloads used to simulate CB rounds, differences in the local environment and data acquisition hardware used, and variations in range and aspect angle of the blasts with respect to sensor emplacement positions. We show that information extracted via different levels of decomposition produce repeatable properties that can be processed and combined to ensure consistently reliable discrimination between HE and CB blasts.

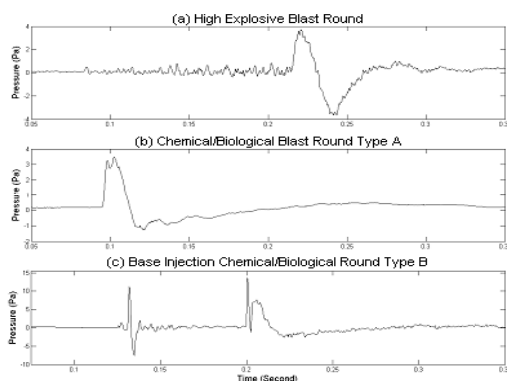


Figure 2 – Acoustic Signatures from (a) HE blast recorded at DPG, (b) Type A Simulated CB Round burst recorded at DPG and (c) Air burst for the Type B White Phosphorus Base Injection round recorded at YPG

2. WAVELET FEATURE EXTRACTION FOR CLASSIFICATION

In this paper, wavelet analysis is used to identify distinct, disjoint feature sets that remain consistent for a given class of round, and do not degrade significantly with long-range propagation. Figure 2 shows the complex, non-stationary response of the signatures to be evaluated, which are categorically poor candidates for feature extraction and segmentation via Fourier analysis or the short-time Fourier Transform. On the other hand, the non-stationary, transient and often oscillatory nature of these signals may be characterized proficiently with wavelet bases that effectively capture the time-frequency distribution of such signal components. Wavelets are better suited for analyzing transient signals, as they are well localized in time and in frequency, and are able to take into account the scale of multiple signal components [10]. To this end, the wavelet transform can provide a scalable time-frequency representation of artillery blast signatures and should uncover details that are not readily found using conventional signal processing techniques. Noise components that are introduced into the blasts waves prior to being received at the sensor are also a serious impediment to isolating features that can be used to efficiently characterize the signal. One useful feature of utilizing wavelets in multiresolution analysis is that this noise can be isolated and removed from the baseband signal of interest; a process referred to as wavelet denoising. Wavelet denoising is optimal in the sense that the process attempts to remove noise components from signal components regardless of the frequency content of the signal, which is by far more proficient than conventional filtering methods that retain baseband signal components and suppress high frequency noise.

2.1 Discrete Wavelet Transform and Multiresolution Analysis

Derived from subband filters the discrete wavelet transform (DWT) is based on a multiresolution decomposition of a signal to give a coarser and coarser approximation to the original signal by removing high frequency detail at each level of decomposition [11]. The wavelet transform, in other words is a multiresolution transform that maps low frequency information of signals into a coarsely sampled subspace and maps high frequency information into more finely sampled subspaces. The DWT is defined by the scaling function

$$\phi(x) = 2^{1/2} \sum_{k=0}^{L-1} h_{k+1} \phi(2x - k) \quad (1)$$

and a wavelet function

$$\psi(x) = 2^{1/2} \sum_{k=0}^{L-1} g_{k+1} \phi(2x - k) \quad (2)$$

where h_k and g_k are analysis filters. In order to retain the characteristics of the transient signals in question choosing the appropriate wavelet filter is important. The choice of the wavelet scaling function should have properties similar to the original signal because the quality of wavelet decomposition depends significantly on the ability to approximate the signal with wavelets. The output of the wavelet transform shows the correlation between the signal and wavelet as a function of time. The easiest method for selecting a wavelet is to copy a signal's time-frequency behavior, however, most transient signals cannot be used as wavelet basis functions since they resemble exponentially damped sinusoids and do not possess a zero mean. The wavelet basis used in this paper is the db5 wavelet defined by Daubechies [9] that has the scaling function and translation function shown in Figure 3(a), and 3(b) respectively.

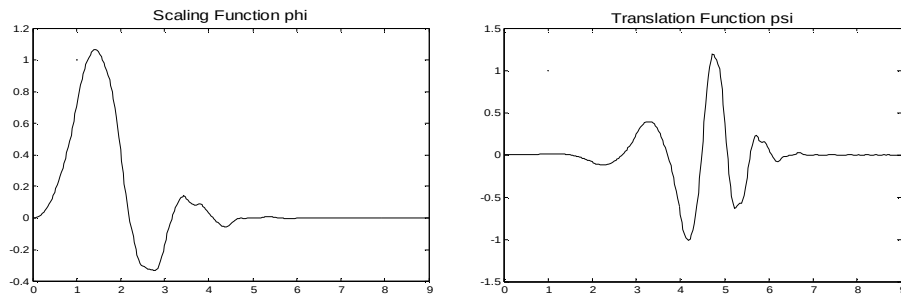


Figure 3 – Daubechies wavelet n=5 (a) Scaling Function (b) Translation Function

Note that scaling function strongly resembles the blast signature for the HE and "Type A CB signals" shown in Figure 3a and 3b, as well as the short pulse preceding the main blast in Figure 2c. Shown in Figure 4, the multiresolution filter bank using low pass and high pass wavelet filters to decompose an input signal into different frequency bands is used to implement the DWT. At each level of decomposition, the high pass filter defined in (2) produces the detail information, D_n , while the low pass filter associated with the scaling function in (1) produces the coarse approximations A_n . Often referred to as the Mallat algorithm a process of successive lowpass and high pass filtering of the input signal is used to implement the DWT [12]. The resulting banks of dyadic multirate filters are used to split up the input signal's frequency components into different subbands at each decomposition level, each with a subset of frequencies spanning half of the original frequency band. If the original signal is sampled at a frequency of f_s Hz, then the output of the first high pass filter which is the first detail coefficient D_1 , captures the band of frequencies between $f_s/2$ and $f_s/4$. This reduces the uncertainty in the frequency by a factor of 2 as the frequency resolution doubles. In the same manner, the highpass filter in the second stage would capture signal components with a bandwidth between $f_s/4$ and $f_s/8$, and so on. In this way, an arbitrary resolution in frequency can be obtained. Since the input signal at each stage of decomposition contains the highest frequency equivalent to twice that of the output stages, it can be sampled at half the original sampling frequency, thus discarding half the samples with no loss of information. This decimation by 2 halves the time resolution of the entire signal as the input signal is represented by half of the total number of samples and effectively doubles the scale.

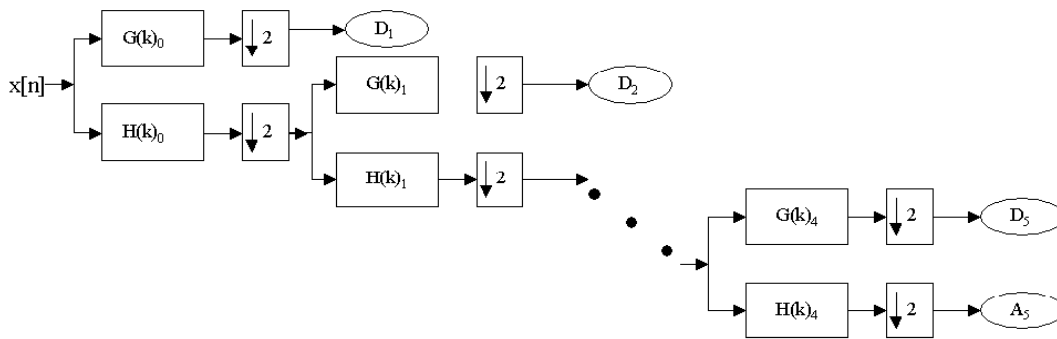


Figure 4 – A Multirate Filter Bank used as a Five Level Wavelet Decomposition Tree

2.2 Wavelet Features for Classification

Figures 5 (a-c) show blast waves from the three unique artillery rounds of interest, followed by four signals derived from the multiresolution decomposition shown in Figure 4; the coefficients at level 5 and detail signals at level 5, 4, and 3 respectively. The proposed feature space is comprised of primitives derived from the normalized energy distributions within the details at level 5, 4, and 3, centered about the maximum value of the blast wave. In addition, values derived from the coefficients at level 5 a fourth feature is obtained. Features such as the rise time for the blast wave and the low frequency content found within the acoustic signals is least attenuated over long propagation distances when compared to some of the predominant features initially identified for discrimination. From the plots in Figure 5, it can be seen that at varying levels of decomposition beyond level 2, differences in the energy distribution within the details differ dramatically prior to the max peak pressure of each blast with respect to energy distributions after the blast has occurred. Furthermore, this energy is not amplitude dependant as the baseband signal is filtered by the scaling function and only the high-frequency noise components are captured within the details after decomposition.

Let t_p denote the sample time where the maximum peak over pressure of a blast wave occurs. Let $t_0 = \alpha * f_s$ designate an integration start time, and $t_F = \beta * f_s$ denote integration stop time. We restrict the values α, β such that $0 < \alpha, \beta \leq 1$ for $t_p > t_0$ and $t_F > t_0$ such that $t_p, t_0, t_F \in I$. Define the energy distribution within the details just prior to the max peak pressure as

$$D_k^- = \frac{1}{N} \sum_{n=t_0}^{t_p} |D_k(n)| \quad (3)$$

where $N = t_p - t_0$. Define the energy distribution at the point immediately after the max pressure occurs as

$$D_k^+ = \frac{1}{M} \sum_{m=t_p}^{t_F} |D_k(m)| \quad (4)$$

where $M = t_F - t_p$. Since $D_k^-, D_k^+ > 0$ we are able to define analytic features $x_{Dk} \in \mathfrak{R}^k$ using the ratio of the normalized energy distributions within the details.

$$x_{Dk} = \log_{10} \left(\frac{D_k^-}{D_k^+} \right) \quad (5)$$

The first 3-tuple of the features space consists of x_{D5}, x_{D4}, x_{D3} which we propose are useful to characterize the artillery blasts of interest. A fourth feature is derived from the wavelet coefficient obtained at level 5. We integrate the magnitude of the area for the coefficients between the start and stop times defined for the details

$$A_{5AREA} = \log_{10} \left(\frac{1}{K} \sum_{k=t_0}^{t_F} |A_5(k)| \right) \quad (6).$$

Figure 6 (a-f) shows all permutations of 2-D subspaces for the entire set of data collected at Dugway Proving Ground that was used to train and initially benchmark the performance of a neural network classifier. An 'x' is used to indicate features derived from the CB blasts while the corresponding HE feature points are depicted with a '.' Note that the resulting subspaces reveal a high degree of separability, notwithstanding the 540 – 1660m variability in range at which the data was acquired.

3. NEURAL NETWORK CLASSIFIER

Neural networks are a powerful tool used to solve difficult classification (mapping) problems with complex decision regions that are often required to ensure separability between classes and a proven ability to realize non-linear discriminant functions.. The use of neural networks for classification is well documented and the requirements for training are well known. The standard multilayer feedforward neural network shown in Figure 7 was chosen due to its ability to learn mappings of any complexity, provided that the number of hidden layer neurons is sufficient to accommodate the number of separable regions that are required to solve a particular classification problem. In general, the network contains N_i inputs, N_h hidden layer neurons and N_o output layer neurons with no interconnections within a single layer. In the three-layer network shown, the connection weight between the i^{th} input and j^{th} hidden layer neuron will be denoted as w_{ij} and v_{jk} will be used to denote the connection weight between the j^{th} neuron in the hidden layer and k^{th} output layer neuron. A bias for the j^{th} hidden layer neuron is denoted as b_j and the bias for the k^{th} output neuron is $\hat{b}_k$. x^P denotes the set of P , N -tuple feature vectors $x_n \in \mathfrak{R}^N$, $x^P = [x_1^P, x_2^P, \dots, x_N^P]$ used to discriminate between the different classes of artillery.

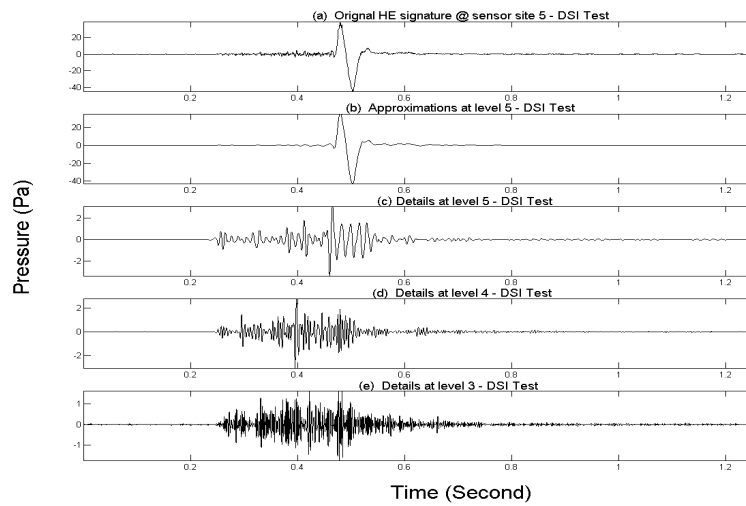


Figure 5a –HE Signal and Wavelet Decomposition to Level 5

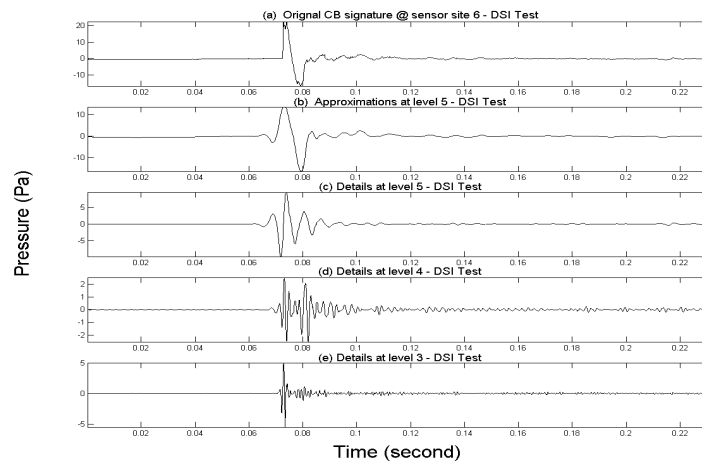


Figure 5b –Type A CB: Original Signal and Wavelet Decomposition to Level 5

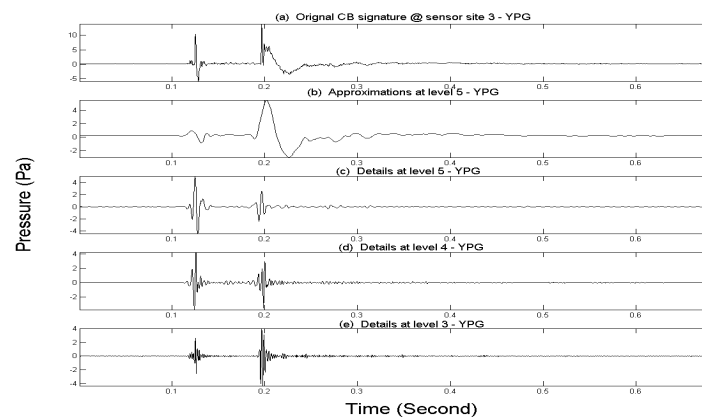


Figure 5c – Type B CB: Original Signal and Wavelet Decomposition to Level 5

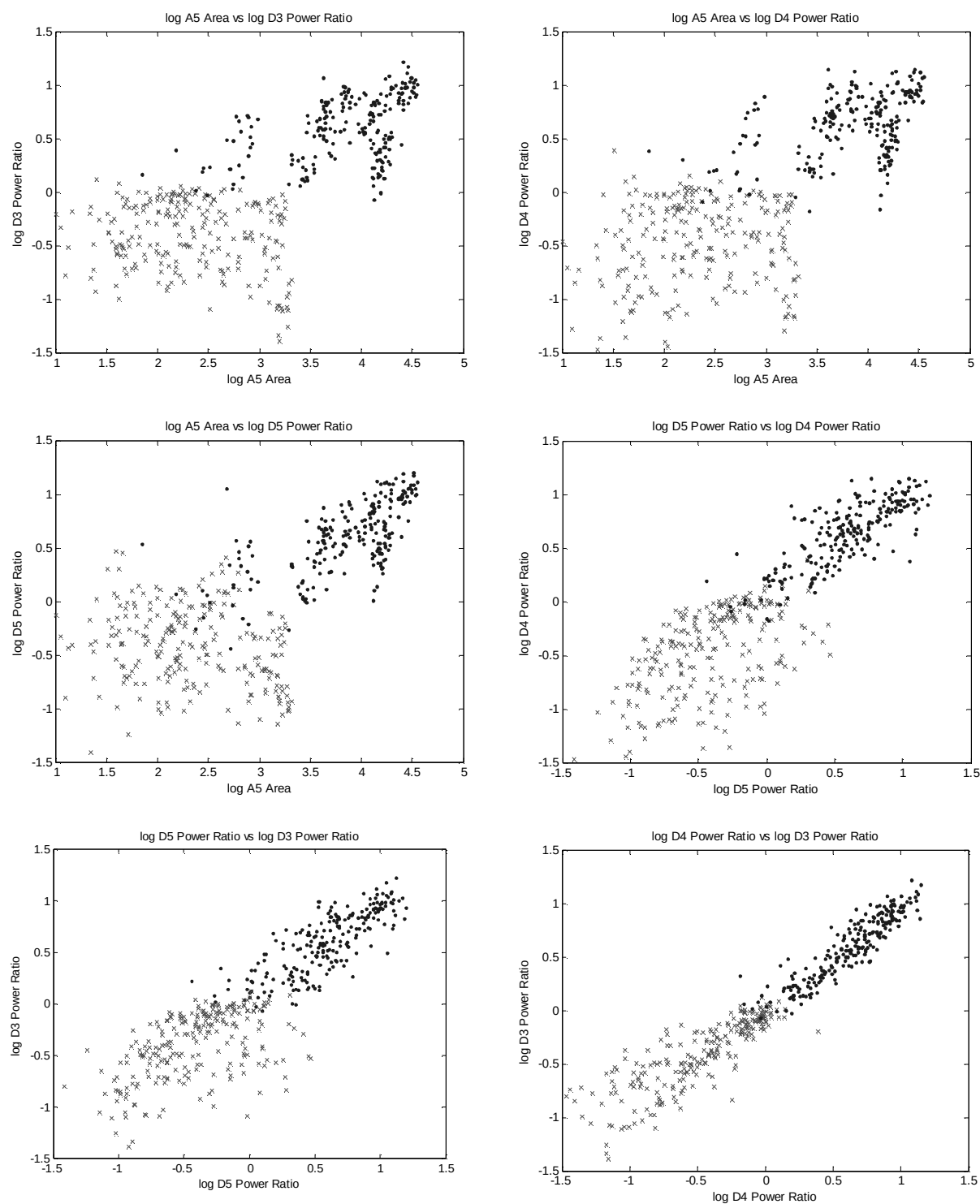


Figure 6 – 2D Feature Spaces derived from the “known” DPG data set (a) A5area vs. xD3 (b) A5area vs. xD4 (c) A5area vs. xD4 (d) xD5 vs. xD4 (e) xD5 vs. xD3 and (f) xD4 vs. xD3

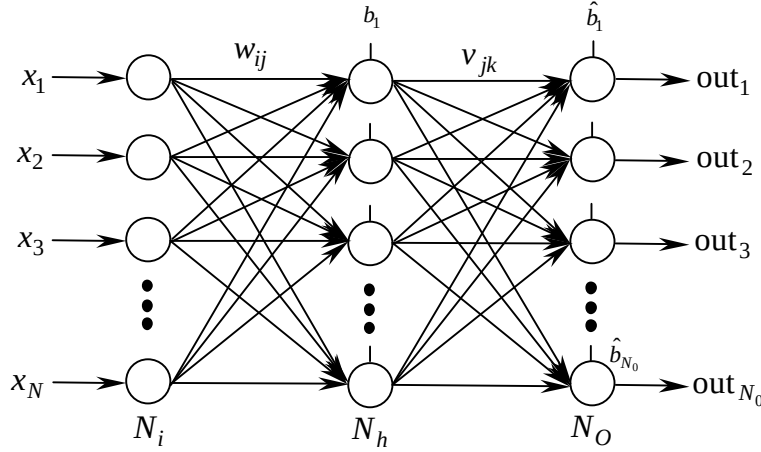


Figure 7– Standard Multilayer Feedforward Neural Network

The input layer units or “neurons” propagate signals to the hidden layer but do not perform any computations. Neurons in the hidden layer and the output layer compute their response by taking a weighted sum signals from the previous layer plus a bias, and then passing the sum through an activation function. For example, the output of the j^{th} neuron in the hidden layer upon the presentation of the p^{th} input pattern x^p is given by $\text{out}_j^p = f(\text{net}_j^p)$ where

$$\text{net}_j^p = \sum_{i=1}^N w_{ij} x_i^p + b_j, \text{ and } f \text{ is an activation function, that was chosen to be the sigmoid function}$$

$$f(\text{net}) = \frac{1}{1 + e^{-\text{net}}} \quad (7)$$

The neural network classifier used to discriminate between HE and CB artillery blasts was trained using the generalized delta rule or back propagation algorithm. The algorithm sequentially adjusts the interconnection weights within the network, subsequent to the application of all patterns in a training set, a routine commonly referred to as an epoch. In general, when an input pattern x^p from a training set is presented to the network, it produces an output that is different from the target value, say d^p . The error for the specific pattern is defined as the squared error $E_p = \frac{1}{2} \sum_{k=1}^{N_o} (d_k^p - \text{out}_k^p)^2$. An unconstrained nonlinear optimization is performed to minimize the total error function of the network

$$E(w) = \sum_{k=1}^P E_p(w) \quad (8)$$

through the incremental computation of the gradient of the error in equation (8), and successive adjustment of the interconnection weights so as to achieve the global minimum error corresponding to $E(w) = 0$.

4. RESULTS

Features were extracted from both data sets using the DWT as described in section 3, and the 4-tuple feature vector $x^p = [x_{D5}^p, x_{D4}^p, x_{D3}^p, A_{5\text{AREA}}^p]$ was constructed according to equations (5) and (6). Experiments were then conducted to measure the separability of the feature space and benchmark the performance of the proposed neural network classifier.

The architecture shown in Figure 7, consisting of a single hidden layer neuron was trained using 22 randomly selected vectors from a total of 461 signatures made available from the data collection exercise at DPG. The training set comprised of 11 samples of HE data and 11 samples of Type A CB rounds collected at each of the 3 sensor sites. The goal of the training process was to obtain a neural network output equal to 0 when the sample features presented to the input layer corresponded to an HE round and an output value of 1 when features from a CB round are applied. The network was trained such that the total error in equation (8) was less than $5e-3$ and a learning rate of 0.1 was used. When tested against the remaining 439 signatures, the networked correctly classified 100% of the remaining 225 CB blasts and misclassified a total of 4 out 214 (or 98.1%) of the remaining HE rounds, for an overall performance of 99.1%. The rounds were classified as CB when the output from the sigmoid activation function in equation (7) was greater than $\frac{1}{2}$ and HE otherwise.

In a subsequent experiment, a neural network consisting of 4 hidden layer neurons was then trained using the entire dataset from the DPG test; a total of 236 Type A CB rounds and 225 HE signatures. The network was then tested against the data collected by NCPA at YPG. The network correctly classified 165 of the total of 166 CB rounds and misclassified a total of 6 out of 57 HE rounds, for a rate of 99.4% and 89.5% respectively. Neural networks consisting of up to 3 hidden layer neurons could not match or surpass the classification performance provided with the architecture comprised of 4 hidden layer neurons. A third experiment was undertaken utilizing the “so-called” blind data from DPG for which truth data was provided by the test facility. The data was tested using a neural network trained with 4 hidden layer neurons and against all the initial DPG data that was considered to be “not blind”. The results obtained for these blasts provided correct classification results of 98.3% of all the CB and 95.7% of all the HE, providing further confidence in the neural network and the results of the classification. However, the same results were obtained when training a neural network comprised of 5 or more hidden layer neurons. Once again, the rounds were classified as CB when the output from the sigmoid activation function (7) produced a value greater than $\frac{1}{2}$ and HE otherwise. A summary of the neural network performance is shown in Table 1 below.

Experiment #	Training Data	Test Data	Classification	Percentage
1	11 CB (DPG)	225 CB (DPG)	225 CB / 0 HE	100%
	11 HE (DPG)	214 HE (DPG)	210 HE / 4 CB	98.1 %
2	236 CB (DPG)	166 CB (YPG)	165 CB / 1 HE	99.4 %
	225 HE (DPG)	57 HE (YPG)	51 HE / 6 CB	89.5 %
3	236 CB (Blind)	230 CB (Blind)	226 CB / 4 HE	98.3 %
	225 HE (Blind)	184 HE (Blind)	174 HE / 8 CB	95.7 %

Table 1 - Summary of Classification performance results

5. CONCLUSION

In this paper, feature extraction methods based on the discrete wavelet transform and multiresolution analysis facilitate the development of a robust classification algorithm that affords reliable discrimination between conventional and simulated chemical/biological artillery rounds via acoustic signals produced during detonation. Amplitude dependant features provide the basis for previously developed discrimination algorithms, which vary dramatically due to signal attenuation and distortion resulting from various propagation effects. Measurements from multiple sensors can be used to obtain range information to a blast under these circumstances, which in turn maintains the potential for a high level of correct classification. In general, it was noted that when the distances from the detonation point to disparate sensor exceeds 1km; the probability of correct classification degrades dramatically, even though visual cues that can be used to exploit class specific attributes are still noticeable within the signature data.

The non-stationary, transient and often oscillatory nature of artillery blast signals may be characterized efficiently with wavelet bases that successfully capture the time-frequency distribution of such signal components. Wavelets are better suited for analyzing transient signals, as they are well localized in time and in frequency, and are able to take into account the scale of discrete signal components. To this end, the wavelet transform has been shown to

provide a scalable time-frequency depiction of blast signatures and has revealed details that are not readily detectable using conventional signal processing techniques. Utilizing the idea of multiresolution analysis provided by the discrete wavelet transform, we were able to break the original signature into subband components thereby removing the higher frequency noise features and creating two sets of coefficients at varying levels of decomposition. These coefficients are obtained each time the signal is passed through a lowpass and highpass filter bank whose impulse response is derived from Daubechies db5 wavelet. Distinct features are obtained through the process of isolating the details of the higher oscillatory components of the signature (i.e. characteristic whining of the shrapnel for HE rounds and weak under pressure for CB rounds). The ratio of the energy contained within the details at varying levels of decomposition is sufficient in discriminating between the vast majorities of artillery blasts.

The discrete wavelet transform has been successfully employed to extract unique, disjoint feature sets that remain consistent for a given class of round and do not degrade dramatically with long-range propagation. Classification results using a neural network trained on such attributes show consistently reliable discrimination of CB artillery rounds that surpasses 98%, even when the network is trained on data acquired during different field test experiments. In addition, the algorithms developed herein allow for the discrimination of base ejection type rounds, which were not part of the original training set used to train the neural network.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the assistance Chris Reiff from the Army Research Lab for his assistance in providing data sets from the Disparate Sensor Integration test at Dugway Proving Grounds, and Amnon Birenzvice and Dave Sickenberger at the Edgewood Chemical and Biological Center.

REFERENCES

1. H.E. Bass, "Aero Acoustic Blast Classification Study (ACBS)", Interim Report for contract 95-F139300-000, January, 1996
2. Birenzvice, A., Sickenberger, D., and Reyes, F., "Disparate Sensor Integration Program Description and Preliminary Data Analysis", Report ECBC-TR-251, Edgewood Chemical Biological Center, September 2002.
3. Birenzvice, A., Redmond, R., Water, P. "High Energy and Chemical Blast Analysis", Report ECBC-TR-310, Edgewood Chemical Biological Center, August 2003.
4. Birenzvice, A., Sickenberger, D., Reyes, F., Reiff, C., Gonski, D., "Force Protection: Early Detection of Chemical/Biological Attack Using Disparate Sensors, ECBC-TR-271, April 2003.
5. Reiff, C. and Gonski, D., "Discrimination of Chemical/Biological vs. High Explosive Artillery Rounds Using Acoustic and Seismic Data Fusion", Poster Presentation, 23rd Army Science Conference, Orlando, Fl. November 2002.
6. Reiff, C., Farques, M., Gonski, D., "Discrimination of Chemical/Biological vs. High Explosive Artillery Rounds Using Acoustic and Seismic Data Fusion", SPIE, Fl. November 2002.
7. Farques, M.P., "Discrimination between Chemical/Biological and High Explosive Seismic Signatures", Naval Postgraduate School, Technical Report NPS-EC-03-002, November 2002.
8. Sickenberger, D., Birenzvice, A., Reyes, F., "Use of Disparate Sensors for the Detection of Chemical and Biological Events", Proc. SPIE conference on Unattended Ground Sensor Technologies and Applications IV, Aug. 2002, pp. 213-224.
9. Daubechies, I., 1992, Ten Lectures on Wavelets: SIAM, Capital City Press.
10. United States Army., 1994, "Chemical and Biological Contamination Avoidance", FM No. 3-3/FM No.11-17 Headquarters Department of the Army, Washington, DC
11. Vetterli, M., Kovacevic, J., "Wavelets and Subband Coding", Prentice Hall, NJ 1995
12. Mallet, S., "A theory for Multiresolution Signal Decomposition: The Wavelet Representation.", IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 11. No. 7. July 1989.